



Prevalence and Associated Factors of Postpartum Depression among Adolescent and Adult Mothers in Khulna region, Bangladesh

Course Title: Masters Level Seminar

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Background

- Postpartum Depression (PPD) prevalence: Globally 17.22%; LMICs 18.6%; Bangladesh 27.75%
- Literature review revealed that, Globally and in LMICs, adolescent mothers' PPD prevalence was higher than adult mothers; however, in Bangladesh adult mothers have higher PPD prevalence.
- From 18 Bangladeshi PPD studies, it was found that 30 factors were riskily and 5 factors were protectively associated with PPD.

Justifications of the Study

- Despite high rate of adolescent childbearing (**93 births/ 1000** teenage girls in **Bangladesh vs. 41.3 births/ 1000** teenage girls **Globally**), limited PPD research was done on this age group.
- Existing Bangladeshi studies have assessed PPD among postpartum mothers; however, none has specifically compared adolescent and adult mothers.
- Factors associated with postpartum depression may vary between adolescent and adult postpartum mothers.

Primary Research Question

- What are the prevalence of PPD among adolescent and adult mothers in Khulna region, Bangladesh?

Secondary Research Question

- What socio-demographic, socio-cultural, and obstetric factors are associated with postpartum depression among adolescent and adult mothers in Khulna region, Bangladesh?

General Objective

- To assess the prevalence and associated factors of PPD among adolescent and adult mothers in Khulna region, Bangladesh

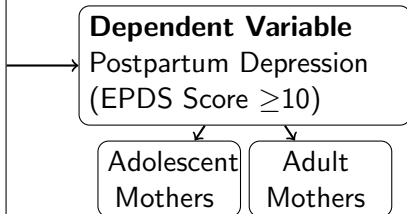
Specific Objectives

- To estimate the prevalence of PPD among adolescent and adult mothers
- To measure the association between participants socio-cultural characteristics and PPD
- To measure the association between participants obstetric characteristics and PPD

Conceptual Framework

Independent Variables

- Sociodemographic variables
 - Maternal age; education; occupation
 - Household income
 - Residence
- Socio-cultural variables
 - Social support (MSSS)
 - Marital dissatisfaction
 - Previous depression history
- Obstetric variables
 - Parity; pregnancy intention
 - Delivery complications; last delivery mode
 - Gender preference



- **Study Design:** Comparative cross-sectional study
- **Study Population:** Adolescent and adult postpartum mothers in Khulna region
- **Study Site:** Khulna Medical College; Gazi Medical College; and Batiaghata Upazilla Health Complex
- **Study Duration:** Between February 2026 and September 2026

- **Inclusion Criteria:**

- Adolescent mothers (age 15-19 years) within 1 year postpartum
- Adult mothers (age 20-49 years) within 1 year postpartum

- **Exclusion Criteria:**

- Mothers having newborns with severe illness
- Mothers having chronic illness
- Mothers having preexisting mental health disorder

Sample Size Calculation

$$\begin{aligned} n &= \frac{r+1}{r} \cdot \frac{\bar{p}(1-\bar{p})(Z_{\beta} + Z_{\alpha/2})^2}{(p_1 - p_2)^2} \\ &= \frac{(1+1)}{1} \cdot \frac{0.33(1-0.33)(1.96+0.84)^2}{(0.26-0.40)^2} \\ &= 177 \text{ per group} \end{aligned}$$

$$\begin{aligned} \text{Total} &= (177 + 177) \\ &= \mathbf{354} \end{aligned}$$

Assumptions

- $p_1 = 0.26$ (Teen)
- $p_2 = 0.40$ (Adult)
- $\bar{p} = \frac{0.26+0.40}{2} = 0.33$
- $Z_{\alpha/2} = 1.96$
- $Z_{\beta} = 0.84$
- $r = 1$

Sampling Technique:

- A **convenience sampling** approach will be used.

Data Collection Tools:

- Bengali version of **Edinburgh Postnatal Depression Scale** (EPDS-B) (10 items)
- **Maternity Social Support Scale** (6 items)
- Socio-demographic questionnaire
- Obstetric questionnaire

Data Analysis Plan

Required Softwares:

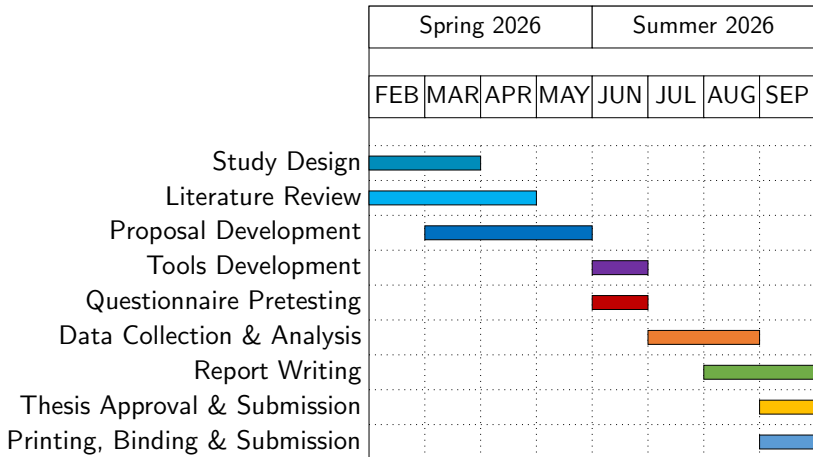
- MS excelsheet for data entry; R Statistical software for data analysis

Required Analysis:

- **Descriptive statistics** to describe PPD prevalence and socio-demographic characteristics
- χ^2 **test** to compare categorical data between adolescent and adult mothers
- **Bivariate logistic regression** to explore the associations between PPD and each of the selected explanatory variables separately for adolescent and adult mothers
- **Multivariable logistic regression** to measure adjusted association between PPD and explanatory variables in the two groups

- Prior to data collection, ethical approval will be obtained from the **Institutional Review Board (IRB) of North South University.**

Study Timeline



Thank You